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Mayank Kumar Singh

Senior Software Engineer

Github
Leetcode
LinkedIn

Backend engineer with 4 years designing, migrating, and operating production systems for JPMorgan Chase's travel platform — from session datastores on AWS Keyspaces to event-driven ingestion with RabbitMQ. Core stack: C#/.NET, PostgreSQL, Redis, AWS. Currently owning backend systems and shipping full-stack features end-to-end at Coursefinder.ai..

Work Experience

Coursefinder.ai — **Nagpur, Maharashtra**
Senior Software Engineer — *10/2025 — Present*

- Optimized PostgreSQL-backed course search by redesigning materialized views and refresh mechanisms, reducing refresh latency by **87%** (**1.5 min** to **20 sec**), improving search freshness, and eliminating recurring CPU bottlenecks.
- Migrated authentication from Auth0 to Kinde with zero downtime, reducing authentication costs by **~60%**.
- Rebuilt platform observability with async logging, centralized log management with S3 archival, and correlation IDs for distributed tracing, enabling end-to-end request tracing across services and replacing manual log hunting during incidents.

JPMorgan Chase & Co. — **Pune, Maharashtra**
Software Engineer II — *02/2025 — 10/2025*

- Led the migration of a session datastore from DSE Cassandra to AWS Keyspaces, implementing dual read/write workflows and data-access optimizations reducing P99 read latency by **25%** on the session-read path serving every travel-platform request, while eliminating operational incidents related to storage and node repairs.
- Developed the Backend-for-Frontend (BFF) layer for the Travel Platform Configuration Portal, consolidating multiple downstream API calls into single orchestrated endpoints, cutting UI data-fetch latency by **40%**.
- Designed and implemented an event-driven hotel content ingestion workflow using RabbitMQ and scheduled jobs, synchronizing supplier hotel static content into Chase-managed databases and reducing hotel search latency by **10%** through local content availability.

Software Engineer I — *07/2022 — 01/2025*

- Built Hotel Dynamic Cross-Sell as an event-driven precompute pipeline — an AWS Lambda consuming cart-add events from Kafka, executing hotel searches for the added item, and caching results in Redis with DynamoDB persistence — enabling instant upsell rendering at checkout and driving an **11%** increase in hotel bookings (A/B tested).
- Drove application log standardization and index segregation initiative, improving system observability, reducing debugging effort by **10%** and accelerating Kibana log retrieval by **20%**.
- Split a shared Redis cluster serving **4** microservices into per-service instances, isolating cache workloads to eliminate cross-service contention, reducing cache read latency by **20%** and removing a shared point of failure.
- Implemented mock support for Content Service dependencies, improving automation pass rates from **88%** to **99%**.

Personal Projects

API Rate Limiter Library | **Github** — **2025**
C# · .NET Core · Redis · ASP.NET Core

- Pluggable strategy architecture supporting Token Bucket, Sliding Window, and Fixed Window algorithms with per-user, per-IP, and per-endpoint policies.
- Distributed rate limiting via Redis with Lua-scripted atomic operations, ensuring correctness under concurrent load.

Skills

Languages	C#, JavaScript, TypeScript, SQL
Libraries and Frameworks	.NET Core, .NET MVC, React.js, Next.js, Node.js, Express.js
Databases	SQL Server, PostgreSQL, Cassandra, MongoDB, Redis
Cloud & Infrastructure	AWS (EC2, S3, Lambda, Aurora, ElastiCache, AWS Keyspaces, DynamoDB, ELB, ASG, Route 53)
Architecture & Messaging	Microservices, REST APIs, Distributed Systems, Event-Driven Architecture, Kafka, RabbitMQ
Monitoring & Observability	Grafana, Kibana
AI-Assisted Development	Claude, Cursor, GitHub Copilot — code review, system design exploration, test generation

Education

B.Tech in Computer Science — **June 2018 – June 2022**
RCC Institute of Information Technology (MAKAUT) — *Kolkata, West Bengal*